

## EDUCATION

## University of Virginia

Ph.D. Student in Systems Engineering

Charlottesville, VA, USA

2026–2031 (*Anticipated*)

- **Advisor:** Prof. Hongru Du
- **Research Interest:** AI for Science, Large Language Model Agents, Complex Systems, Computational Social Science, Social Stratification, Social Networks

## Hong Kong Baptist University

B.S.S.C. (Hons.) in Sociology, Minor in Computer Science

Hong Kong SAR, China

2022–2026

- **Classification:** First Class Honors
- **GPA:** cGPA: 3.63/4.0, Major: 3.70/4.0, Minor: 3.52/4.0
- **Relevant Coursework:** Intro to AI & ML (A-) | NLP & LLM (A) | Social Stat. (A) | Intro to Python & its Applications (A-) | Problem Solving Using OOP (A) | Calc., Prob. & Stat. for CS (A-) | Sociological Classical Theory (A) | Sociological Theory II (A-) | Social Networks (A-)
- **Honors Project:** How Family Socioeconomic Status Shapes Chinese Graduates' School-to-Work Transition: The Roles of University Experiences and Social Capital

## PUBLICATIONS

1. [Under Review] **Haoyang Li**, Runzhou Liu, Yao Li, Amy Wesolowski, Sen Pei, Hongru Du<sup>✉</sup>. (2026). Theory-informed generative agents for modeling human mobility. *Preprint available at Research Square*.
2. [Under Review] Runzhou Liu, Claire Jong, **Haoyang Li**, Yiming Cao, Qing Yao, Teresa Yamana, Sen Pei<sup>✉</sup>, Hongru Du<sup>✉</sup>. (2026). Simulating population compliance with pandemic interventions using Large Language Models. *medRxiv* 2026.05.12.26352942.
3. [Under Review] Qingbin Zeng, Ruotong Zhao, Jinzhu Mao, **Haoyang Li**, Fengli Xu<sup>✉</sup>, & Yong Li. (2025). CrimeMind: Simulating urban crime with multi-modal LLM agents. *arXiv preprint arXiv:2506.05981*.
4. [Working Paper] **Haoyang Li**, Xiao Jia, & Zhanzhan Zhao<sup>✉</sup>. (2025). The emergence of altruism in Large-Language-Model agents society. *arXiv preprint arXiv:2509.22537*.
5. [Working Paper] **Haoyang Li**, Songwei Li, Xiao Liu, Bill Shi, Fengli Xu<sup>✉</sup>, & James Evans<sup>✉</sup>. (2025). Dispersed knowledge, decentralized decisions: A labor market for multi-agent task allocation. *Poster presented at The 1<sup>st</sup> International Conference on AI Scientists (ICAIS 2025)*. [Poster]

RESEARCH  
EXPERIENCE

- **Graduate Research Assistant** | Du Lab, University of Virginia 2026.08 – Present
- **Research Intern** 2025.07 – 2026.07
  - Supervised by **Prof. Hongru Du**, Assistant Professor, Department of Systems and Information Engineering, University of Virginia
  - *Research Focus:* AI for Science, Modeling Human Behavior through LLM Agents
- **Research Intern** | Knowledge Lab, University of Chicago 2025.10 – 2026.08
  - Supervised by **Prof. James Evans**, Max Palevsky Professor, Department of Sociology, University of Chicago
  - *Research Focus:* AI Scientists for Market Allocation
- **Research Intern** | Future Intelligence Lab, Tsinghua University 2025.03 – 2026.08
  - Supervised by **Prof. Fengli Xu**, Assistant Professor, Department of Electronic Engineering, Tsinghua University
  - *Research Focus:* LLM Agents, Social Simulation

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| RESEARCH<br>EXPERIENCE<br>(CONT'D)                                                                                                                               | <b>Research Assistant</b>   School of Artificial Intelligence, <b>The Chinese University of Hong Kong, Shenzhen</b> 2025.07 – 2025.11                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|                                                                                                                                                                  | <ul style="list-style-type: none"><li>Supervised by <b>Prof. Zhanzhan Zhao</b>, Assistant Professor, School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen</li><li><i>Research Focus</i>: LLM and Game Theory</li></ul>                                                                                                                                                                                                                                                                                                                                                             |  |
|                                                                                                                                                                  | <b>Research Assistant</b>   Department of Sociology, <b>Hong Kong Baptist University</b> 2024.03 – 2024.12                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|                                                                                                                                                                  | <ul style="list-style-type: none"><li>Supervised by <b>Prof. Yinni Peng</b>, Associate Professor, Department of Sociology, Hong Kong Baptist University</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|                                                                                                                                                                  | <b>Research Assistant</b>   Department of Sociology, <b>Hong Kong Baptist University</b> 2024.03 – 2024.06                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
| <ul style="list-style-type: none"><li>Supervised by <b>Prof. Jin Jiang</b>, Associate Professor, Department of Sociology, Hong Kong Baptist University</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| AWARDS<br>AND<br>HONORS                                                                                                                                          | <ul style="list-style-type: none"><li><b>Halloran Scholarship</b>, 18<sup>th</sup> SIS MID, Emory University (Full Reg. Waiver)2026.03</li><li><b>President’s Honor Roll</b>, Hong Kong Baptist University<br/>Semester 1, 2022–2023; Semester 1&amp;2, 2023–2024;<br/>Semester 1&amp;2, 2024–2025; Semester 1, 2025–2026</li><li><b>Dean’s List</b>, Hong Kong Baptist University<br/>Semester 2, 2022–2023; Semester 2, 2025–2026</li><li><b>HKSAR Government Talent Development Scholarship</b>, HK\$10,0002023.04</li></ul>                                                                                |  |
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| TALKS                                                                                                                                                            | <ul style="list-style-type: none"><li><b>Mainland Undergraduate Admissions Talk</b>, Hong Kong Baptist University2026.05</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
| TEACHING                                                                                                                                                         | <ul style="list-style-type: none"><li><b>Teaching Assistant</b>, Hong Kong Baptist University2024.01 – 2024.04<br/>Course: COMP1005 Essence of Computing • <a href="#">[Course Outline]</a></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
| SKILLS                                                                                                                                                           | <ul style="list-style-type: none"><li><b>Languages</b>: Chinese Mandarin (Native), Chinese Cantonese (Listening), English (Working Language), <math>\LaTeX</math></li><li><b>Programming &amp; Software</b>: Python, Java, C, SQLite, MATLAB, Stata, SPSS, Microsoft Office</li><li><b>Interests</b>:<ul style="list-style-type: none"><li><b>Chinese Writing</b>: won the <i>First Prize</i> of the 24<sup>th</sup> National New Concept Writing Contest</li><li><b>Hiking</b>: I left many footprints along the <i>MacLehose Trail</i> and other mountains across Hong Kong and Shenzhen</li></ul></li></ul> |  |